

EDUCATION

University of Illinois at Urbana-Champaign

Master Science in Information Management GPA 4.0/4.0

Illinois, US

Aug. 2024 – Present

ShanghaiTech University

Bachelor of Engineering in Computer Science

Shanghai, CN

Sep. 2020 – June 2024

RESEARCH INTERESTS

My research interests lie primarily in **Large Language Models (LLMs)**, **Vision-Language Models (VLMs)**, autonomous **LLM Agents** and **AI4Science**.

PUBLICATIONS

[1] Pick Your Textual Gradients

Yifan Xu, Yixuan Li, Xinzhuo Li, Lijun Yu, Haohan Wang, under review, 2025.

[2] Tracing LLM Reasoning Processes with Strategic Games: A Framework for Planning, Revision, and Resource-Constrained Decision Making

Xiaopeng Yuan, Xingjian Zhang, Ke Xu, Yifan Xu, Lijun Yu, Jindong Wang, Yushun Dong, and Haohan Wang, arXiv preprint arXiv:2506.12012, 2025.

[3] NanoNeRF: Robot-assisted Nanoscale 360° reconstruction with neural radiance field under scanning electron microscope

Xiang Fu, Yifan Xu, Hu Su, Song Liu, in IROS(Oral), Abu Dhabi, Oct. 2024.*

[4] A Review of Deep-learning-based Super-Resolution: from methods to applications

Ying Li, Yifan Xu, Hu Su, and Song Liu, in Pattern Recognition, paper id: 10.1016/j.patcog.2024.110935.

RESEARCH EXPERIENCE

University of Illinois at Urbana-Champaign

Graduate Research Intern | Advisor: Prof. James M. Rehg & Prof. Haohan Wang

Illinois, US

Feb. 2025 – Present

Aligning VLM Abstract Reasoning with Humans

Oct. 2025 – Present

- Aligning VLM abstract reasoning with human cognition by leveraging SFT and GRPO, with a specific focus on evaluating out-of-distribution (OOD) performance.

Pick Your Textual Gradients

Jun. 2025 – Sep. 2025

- Developed a novel framework to stabilize automated prompt optimization by introducing two core mechanisms: Error-Driven Refinement, which generates high-quality learning signals exclusively from incorrect model outputs, and Regularized Verification, which prevents over-specialization and performance degradation on general tasks.

Tracing LLM Reasoning Processes with Strategic Games

Feb. 2025 – May. 2025

- Designed and developed the "Turn-based Attribute Game" for the AdvGameBench framework, a complex environment with elemental attributes and skill budgets to test the strategic planning and decision consistency of LLMs.
- Evaluated 12 state-of-the-art LLMs within the game, analyzing process-level metrics like Correction Success Rate (CSR) and Over-Budget Rate (OBR) to quantify their ability to plan, revise, and manage resources under constraints.

ShanghaiTech University

Undergraduate Research Intern | Advisor: Prof. Song Liu

Shanghai, CN

Feb. 2023 – Jan. 2024

NanoNeRF: Nanoscale 360° reconstruction with NeRF under SEM

Oct. 2023 – Jan. 2024

- Invented new super resolution and 3D reconstruction techniques to enhance the precision and speed of controlling nanorobots, typically observed under a Scanning Electron Microscope (SEM), where imaging quality and speed are inversely related.

- Achieved rapid reconstruction of nanotubes captured under SEM by modifying the Neural Radiance Field’s pinhole imaging model to parallel projection using Pytorch, aligning it with the physical imaging process of SEM, improving average PSNR from 22.3 to 28.13.

Review of Deep-learning-based Super-Resolution

Feb. 2023 – Sep. 2023

- Conducted extensive research on blind and non-blind super-resolution methods, from classification to application scenarios, and authored a comprehensive review paper.

TECHNICAL SKILLS

Languages: Python, Java, HTML/CSS, TypeScript, JavaScript, SQL, Markdown, Shell, R, MATLAB

Developer Tools: Docker, AWS, VS Code, Linux/Unix, MySQL, Microsoft SQL, MongoDB, PostgreSQL

Technologies/Frameworks: Pytorch, Tensorflow, Numpy, Pandas, Ipywidgets, Matplotlib, Anaconda, OpenAI, Git

INDUSTRY EXPERIENCE

Tektronix, Inc

Shanghai, China

Digital Transformation Intern

Sep. 2023 – Jan. 2024

- Developed a supplier management application using Power Apps, enabling finance staff to register, add, delete and find suppliers, and increased approval efficiency by automating the submission of supplier approval requests and reminders to finance staff, standardizing form formats.
- Developed RESTful APIs using Power Automate to manage CRUD operations for supplier data and transmit processed data to downstream financial services for final processing, leveraging SharePoint and Microsoft SQL for data storage.
- Developed a system using Python to automate the processing of daily bank receipt emails for the finance department by fetching receipt data from Outlook email attachments, removing unnecessary fields, and transforming it into Excel format.

OPEN SOURCE EXPERIENCE

VizSeed - Domain Specific Language for VChart and VTable | TypeScript

June 2025 - Present

- Designed and implemented a domain-specific language (DSL) that abstracts complex multi-dimensional data visualization into declarative syntax, reducing chart configuration complexity by 80% and enabling seamless chart type switching (bar, line, pie, table) without rewriting specifications.
- Architected a data-driven visualization pipeline that transforms raw datasets into standardized VizSeed format through dimension-measure abstraction, supporting arbitrary dimensional data visualization and cross-platform chart rendering for both VChart and VTable engines.

@VisActor/VTable | TypeScript, JavaScript

May 2025

- Developed dynamic data aggregation functionality for PivotTable component that automatically recalculates totals and subtotals when cell aggregators are updated through the in-cell editor.
- Delivered comprehensive testing and documentation by writing unit test suites covering edge cases for aggregation updates, creating **interactive official demos** showcasing the new PivotTable editing capabilities, and authoring technical documentation for developer integration and API usage.

TEACHING EXPERIENCE

CS100: Introduction to Programming

Shanghai, CN

Teaching Assistant

Jan. 2023 - Jun. 2023

- Hosted weekly office hours to tutor students on C/C++ programming concepts and assist with debugging assignments.
- Graded programming assignments, quizzes, and exams, providing constructive feedback.